USHA S. REDDY

Final-Year B.E. AI & ML Student | Machine Learning | Deep Learning | Artificial Intelligence | Python
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Portfolio: https://usha-s-reddy-portfolio.vercel.app

SUMMARY

Final-year B.E. student (AI & ML, 8.16 CGPA, graduating 2027) skilled in Python, SQL, data analytics, exploratory data analysis (EDA), machine learning, and the basics of deep learning. Completed four AI/ML internships and built projects in computer vision, deep learning, transformers/LLMs, and RAG. Seeking a Machine Learning or AI internship or entry-level role to build and deploy machine learning and AI solutions.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, HTML, CSS
Data Analytics: Data Analysis, Exploratory Data Analysis (EDA), NumPy, Pandas, Power BI (basics)
Machine Learning: Scikit-learn, OpenCV, Supervised and Unsupervised Learning, Deep Learning (basics)
AI and LLMs: Transformers and LLMs (basics), RAG (basics), Prompt Engineering (basics), IBM watsonx.ai, IBM Granite models
Tools: Git/GitHub, Docker (basics)
Core CS (foundational): DBMS, Operating Systems, OOP concepts, Time and Space Complexity

INTERNSHIP EXPERIENCE

Unlox | Machine Learning Intern (Hybrid) | Bangalore *Aug 2026 – Present*
- Built the capstone ML project, a self-learning Profile-Based Matching Algorithm (see Projects).

Prodigy InfoTech | Generative AI Intern | Remote Internship *Aug 2026 – Sep 2026*
- Completed five AI projects: GPT-2 text generation and fine-tuning, pix2pix image-to-image translation, neural style transfer, Markov chain text generation, and pre-trained image generation.

Infosys Springboard | AI Intern | Virtual Internship *Jul 2026 – Sep 2026*
- Built the Consumer Attention Mapping System, an AI retail analytics project, following a 4-milestone, 8-week plan.

IBM SkillsBuild x EduNet Foundation | AI and Cloud Intern | Virtual Internship *12 Jun 2026 – 10 Jul 2026*
- Completed a program on quantum computing fundamentals, prompt engineering, and agentic AI; built a RAG-based agent (see Projects).

PROJECTS

Unlox | **Profile-Based Matching Algorithm** *Python, NLTK, scikit-learn, TF-IDF, Pandas, NumPy*
- Hybrid recommender matching people on profile text, MBTI type, and location, and learning from every Accept/Reject.
- TF-IDF text similarity, MBTI compatibility matrix, and haversine location score; per-user logistic regression learns weights.
- Top-5 acceptance 70% to 87% and hold-out ROC-AUC 0.834 to 0.871 on synthetic data; interactive Colab demo.

Infosys Springboard | **Consumer Attention Mapping System (CAMS)** *Python, YOLOv8, OpenCV, DeepSORT/ByteTrack, FastAPI*
- AI retail analytics system that converts store surveillance video into insights on what shoppers notice and engage with.
- Detects and tracks shoppers using YOLOv8 with DeepSORT/ByteTrack, then analyzes gaze, dwell time, and product interaction.
- Produces attention heatmaps and a weighted product-attractiveness score to rank products by shopper attention.

IBM SkillsBuild | **AI-Powered LaTeX Diagram Generator** *IBM watsonx.ai, Granite models, LangFlow*
- RAG-based AI agent that turns plain-English requests and sketches into publication-ready LaTeX/TikZ diagrams for academic work.
- Retrieves context from a vector-indexed knowledge base to generate flowcharts, UML, trees, and state machines.
- Gives real-time feedback and accepts plain-English refinement commands to adjust a diagram.

Prodigy InfoTech | **GPT-2 Text Generation and Fine-Tuning** *Python, Hugging Face Transformers, PyTorch*
- Text-generation pipeline built on the GPT-2 transformer language model and adapted to custom text.
- Implements greedy, beam-search, top-k, and top-p (nucleus) decoding strategies.
- Fine-tuned GPT-2 on a custom corpus to produce domain-specific style and vocabulary.

Prodigy InfoTech | **Image-to-Image Translation with cGAN (pix2pix)** *Python, TensorFlow*
- Deep learning model that translates one type of image into another, such as satellite photos to maps.
- Conditional GAN with a U-Net generator and PatchGAN discriminator for paired translation (satellite-to-map, sketch-to-photo).
- Trained end-to-end with combined adversarial and L1 reconstruction loss.

EDUCATION

Sri Siddhartha Institute of Technology | B.E., Artificial Intelligence & Machine Learning | Tumkur *2023 – 2027 (Expected)*
CGPA: 8.16
Vidyanidhi PU College | PUC (12th, PCMB) | Tumkur *2023*

CERTIFICATIONS

AWS Cloud Practitioner Essentials | Power BI (Microsoft, Simplilearn) | AWS Foundations of Prompt Engineering | IBM Getting Started With AI | Introduction to Deep Learning | Infosys Springboard Introduction to NLP | IBM Exploring Quantum Computing

LEADERSHIP AND MEMBERSHIPS

- **Class Representative (CR), 1 year:** served as the communication link between students and faculty.
- **Team Lead:** organized tasks and kept project teams on schedule through to delivery.
- IEEE Student Member